

HotWire — AEC Reviewer Guide

WOOT '26 Artifact Evaluation

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This is a one-page guide for the WOOT '26 Artifact Evaluation Committee.

Artifact location

- **Public source repository:** <https://github.com/sickcell6000/HotWire>
- **License:** GPL-3.0
- **Zenodo DOI:** 10.5281/zenodo.19986377 (snapshot of the `woot26-artifact-rc1` tag; a separate post-AE version will be archived for the camera-ready paper per the WOOT '26 CFA two-stage process).

Badges requested

We request evaluation for two badges:

- **Artifact Available.** *The artifact is publicly available.* Public GitHub repository under GPL-3.0, archived under a release tag, and mirrored to Zenodo DOI 10.5281/zenodo.19986377.
- **Artifact Functional.** *The core functionality of the artifact can be confirmed by the AEC.* One command — `bash verify_artifact.sh` — runs eight checks (F0–F6) in approximately five minutes on a clean Linux / macOS / Windows machine and exits with `8/8 ✓ ALL PASSED`.

The paper's §6 field-test results were collected on 4 production EVs and 7 commercial DC fast-charging stations and rely on physical access to that fleet. The frozen capture bundles in `datasets/real_hw_traces/` preserve the wire-level evidence from those field tests, but reviewers are not expected to re-run them.

What you need

- A Linux, macOS, or Windows machine (any of those work).
- Python **3.11** or newer.
- ~500 MB free disk, ~5 minutes for the headline check.
- **No hardware required.** No internet required after `pip install`.
- **Docker is optional.** `verify_artifact.sh` F1 prefers Docker CI if a daemon is running (278 tests in-container) and transparently falls back to host `pytest` if not (179 tests; 99 Docker-only ones auto-skip). Either way F1 = PASS and the run reaches `8 / 8 ✓ ALL PASSED`.

Three verification paths

You only need to run **one** of these for the Functional badge. Do all three if you want to also see the attacks live and inspect the real-hardware traces.

Path 1 — One-shot functional check (5 minutes)

Step 0 (all platforms): download the artifact zip from Zenodo The “Available” snapshot for this AE submission lives at the Zenodo record below — please download the zip from there rather than `git clone`-ing the repo, so reviewers and authors look at exactly the same bytes:

Download: <https://zenodo.org/records/19986377> (DOI 10.5281/zenodo.19986377)

The download is a single file, `HotWire-woot26-artifact-rc1.zip` (~5 MB). Save it somewhere convenient (e.g. `~/Downloads/`), then follow the platform-specific block below — each starts with `unzip` + `cd` into the unpacked directory.

Linux / Raspberry Pi

```
# One-time system prerequisites (Pi/aarch64 has no PyQt5 wheel,  
# so we use the apt-packaged Qt5 binary and let venv see it):  
sudo apt update  
sudo apt install -y python3-venv python3-dev python3-pip python3-pyqt5 python3-pyqt5.qtsvg unzip  
  
unzip HotWire-woot26-artifact-rc1.zip  
cd HotWire-woot26-artifact-rc1  
python3 -m venv --system-site-packages hotwire-venv  
source hotwire-venv/bin/activate  
pip install -r requirements.txt  
bash verify_artifact.sh
```

macOS

```
unzip HotWire-woot26-artifact-rc1.zip  
cd HotWire-woot26-artifact-rc1  
python3 -m venv hotwire-venv  
source hotwire-venv/bin/activate  
pip install -r requirements.txt  
bash verify_artifact.sh
```

Windows `verify_artifact.sh` is bash-only by design (matches the convention used by every WOOT 2025 accepted artifact). Confirm you have a bash environment first — Start menu → search Git Bash:

- **Git Bash present** (Git for Windows installed): open Git Bash and follow the *Git Bash* block below.
- **Only WSL**: WSL is Linux — follow the **Linux / Raspberry Pi** block above, run inside `wsl`.
- **Neither**: install Git for Windows (~50 MB; bundles Git Bash). Or install WSL.

Windows — Git Bash / MSYS2

```
# Native Windows Python; venv layout is `Scripts/`, not `bin/`.
# (Git Bash includes `unzip`. Alternatively right-click the zip in
# Explorer → "Extract All..." and skip the unzip line.)
unzip HotWire-woot26-artifact-rc1.zip
cd HotWire-woot26-artifact-rc1
python -m venv hotwire-venv
source hotwire-venv/Scripts/activate
pip install -r requirements.txt
bash verify_artifact.sh
```

PowerShell users: swap the activate line for `.\hotwire-venv\Scripts\Activate.ps1`. `verify_artifact.sh` itself still needs Git Bash or WSL because the script is bash-only.

For real-hardware (PLC modem) work, additionally install `libpcap-dev` and `pip install -r requirements-hw.txt`.

Expected last line:

8/8 ✓ ALL PASSED

The script runs eight checks (F0 through F6) covering:

- **F0** OpenV2G codec binary check — auto-builds from `vendor/OpenV2Gx` if the platform binary is missing.
- **F1** pytest regression — 278 unit + integration tests via Docker CI when available, ~179 via host fallback (the Docker-only tests auto-skip).
- **F2** Simulation-mode full V2G session on `::1` loopback (~25 s end-to-end), expects 13 PEV state transitions and CurrentDemand cycles.
- **F3** Parametric matrix: 3 voltages × 3 durations = 9 successful sim-mode sessions.
- **F4** Attack code present and syntactically valid — `py_compile` for both `autocharge_impersonation.py` (A1) and `forced_discharge.py` (A2).
- **F5** Sim-mode attack reach — runs three end-to-end scenarios (A1 EVCCID injection, A2 forged PreCharge voltage, A1+A2 concurrent) and asserts the fabricated values land on the wire.
- **F6** Real-hardware evidence bundles — validates the 11 frozen pcap files in `datasets/real_hw_traces/` open as pcap v2.4.

If any check fails, the script prints the failing command and exits with a non-zero code — please paste that output into an AEC issue.

Path 2 — Interactive GUI demo (10 minutes)

In two terminals on the same machine:

```
# Terminal 1 – rogue charging station
python scripts/run_gui.py --mode evse --sim

# Terminal 2 – rogue vehicle
python scripts/run_gui.py --mode pev --sim
```

Both processes loop back over a virtual PLC link (no Ethernet required). The GUI shows every DIN 70121 message in a four-tab tree (Combined / Rx / Tx / Trace log) and lets you pause-and-override fields before they go on the wire.

To reproduce the two attacks:

1. Open `config/attack_presets.json` to see the 16 bundled presets.
2. In the EVSE GUI, click **Launch attack** → **A1** (EVCCID impersonation) or **A2** (forced-discharge). The PEV side will honor the forged values and the trace log will show the abuse path the paper describes.

Path 3 — Inspect frozen real-hardware evidence (15 minutes)

```
datasets/real_hw_traces/comprehensive_matrix/
├─ test1_a1_evccid/           ← A1 attack on real charging stack
├─ test2_a2_voltage/         ← A2 attack on real charging stack
├─ test3_multi_pause/        ← multi-pause field manipulation
├─ test4_abort/ ... test11_long_running/
├─ FINDINGS.md               ← what each bundle proves
└─ README.md                 ← per-bundle rationale
```

Each bundle has:

- `*.pcap` — raw HomePlug AV / V2GTP capture (open in Wireshark with the HomePlug AV dissector).
- `session.jsonl` — one decoded DIN 70121 message per line, including a `values:` field with the actual EVCCID, voltages, SoC, etc. **No EXI decoder is needed** — the recorded fields are plain JSON.
- `config.json` — exact run parameters (mode, target voltage, attack preset).

The mapping from each bundle to a specific paper claim is in `docs/PAPER_VALIDATION.md` (§ “Suite 0 through Suite 6”).

What maps to what in the paper

Paper claim	Where to verify
§3 DIN 70121 / ISO 15118-2 FSM	F1 + F2 + F3 / <code>hotwire/fsm/fsm_evse.py,fsm_pev.py</code>
§4 A1 EVCCID-impersonation attack	F4 / GUI demo / <code>test1_a1_evccid/</code>
§4 A2 forced-discharge attack	F4 / GUI demo / <code>test2_a2_voltage/</code>
§5 Tooling (FSM, GUI, pause/override)	F0–F2 / GUI demo
§6 Field results on real fleet	Frozen captures only — physical fleet not in scope

Known-good environments

We have run `verify_artifact.sh` to **8/8 PASS** on all of:

- Raspberry Pi OS Bookworm (aarch64) + Python 3.11 + Docker.
- Ubuntu 24.04 LTS (amd64) + Python 3.12 (host pytest fallback).
- Ubuntu 26 “resolute” (amd64) + Python 3.14 (host pytest fallback).
- Windows 11 22H2 + Python 3.12 + Git Bash (MINGW64) + Docker Desktop.